SOF103 C and C++ Programming
Lab Exercise 1
IDE Setup, Output Statement and Escape Sequences

PART A: REVIEW QUESTIONS

1. What are the outputs of the following program segments?

(a)

```
cout << “I am not going to “;
cout << “ school tomorrow morning\n”;
cout << “because I need more rest “
      << “ after the long flight! << Sign”;
```

(b)

```
cout << “\tI am who I am\n\“Mind your\nown business\“\n“
      << “Do not speak so loud!” << endl
      << “\t\\Keep Silence\\ “;
```

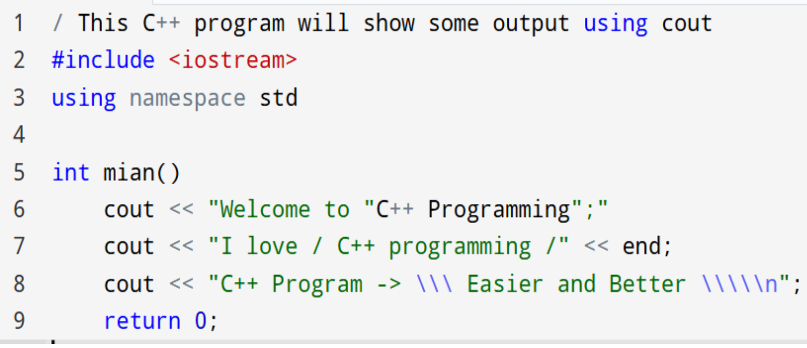
(c)

```
cout << “\t . . . . . . . .\n . . . . \t . . . . \n\t . . . \n\\ . . . . . . . . . . \n\\ . . . . . . . . . .  “;
```

(d)

```
cout << “\\\\- - -////++++++++\n\t//xxxxxx\n\t\\\\xxxxxxxxxx\\“
      << “\n%%%%%%%\t????????\n”
      << “\“9999999999\”\n88888”;
```

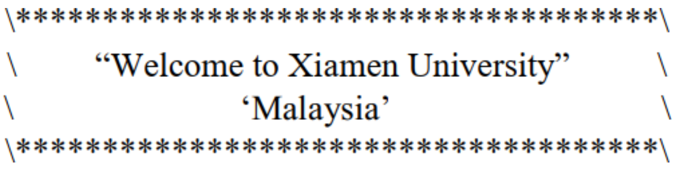
2. Identify the statement that consists of **errors** in the following program segments by rewriting the errors into the correct statement.



```
/ This C++ program will show some output using cout
#include <iostream>
using namespace std

int mian()
    cout << "Welcome to "C++ Programming";"
    cout << "I love / C++ programming /" << end;
    cout << "C++ Program -> \\\ Easier and Better \\\\\n";
    return 0;
```

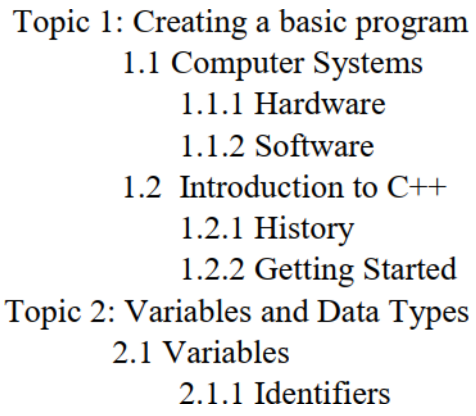
PART B: PROGRAMMING EXERCISES

1. Write a program to print out the following output:



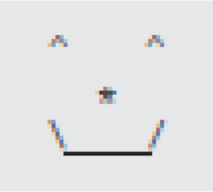
```
\*************************************\
\        “Welcome to Xiamen University”        \
\                      ‘Malaysia’                      \
\*************************************\
```

2. Print the following text in the output window of a C++ program. Take care of the locations of newlines, spaces and punctuations.



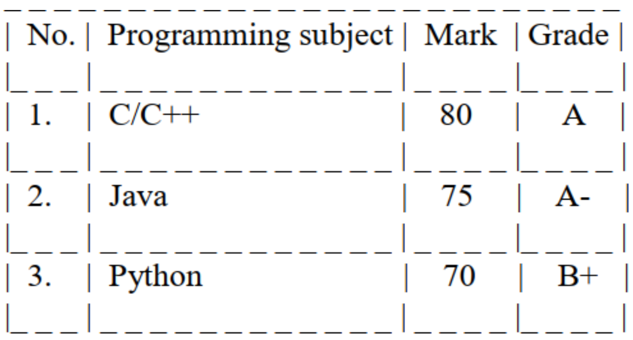
```
Topic 1: Creating a basic program
            1.1 Computer Systems
                  1.1.1 Hardware
                  1.1.2 Software
            1.2  Introduction to C++
                  1.2.1 History
                  1.2.2 Getting Started
Topic 2: Variables and Data Types
            2.1 Variables
                  2.1.1 Identifiers
```

3. Write a program to display the following smiley on the screen.



```
  ^     ^
     *
  \_____/
```

4. Write a C++ program to construct a table as shown below.



```
 _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
| No. | Programming subject | Mark | Grade |
|_ _ _|_ _ _ _ _ _ _ _ _ _ _ _ _|_ _ _ _ _|_ _ _ _ _|
| 1.  | C/C++               |  80  |   A   |
|_ _ _|_ _ _ _ _ _ _ _ _ _ _ _ _|_ _ _ _ _|_ _ _ _ _|
| 2.  | Java                |  75  |   A-  |
|_ _ _|_ _ _ _ _ _ _ _ _ _ _ _ _|_ _ _ _ _|_ _ _ _ _|
| 3.  | Python              |  70  |   B+  |
|_ _ _|_ _ _ _ _ _ _ _ _ _ _ _ _|_ _ _ _ _|_ _ _ _ _|
```

5. Write a C++ program that will produce the following output:

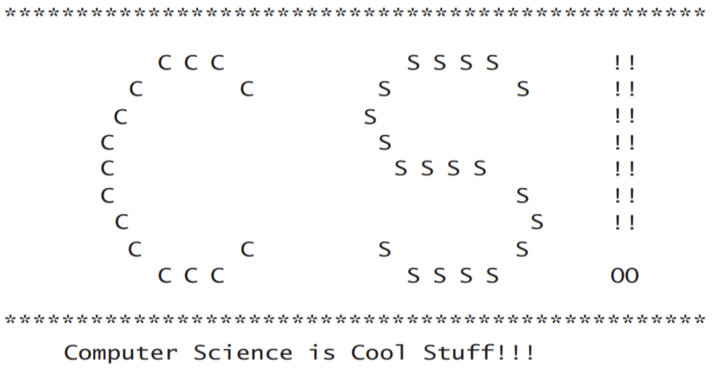
(a)

```
Welcome to C++
https://www.youtube.com

\\\ We’re going to class now \\\


Lecturer says “Programming is fun”
```

(b)

```
Fruit          Price
=====          =====
Apple          RM10
Orange         RM15
Pear           RM20
================
Total          RM45
```

6. Write a program that prints out “C S !” in large block letters inside a border of *s followed by two blank lines then the message “Computer Science is Cool Stuff”. The output should look as follows:



```
**************************************************

            C C C              S S S S         !!
          C       C          S         S       !!
         C                  S                  !!
        C                    S                 !!
        C                     S S S S          !!
        C                              S       !!
         C                              S      !!
          C       C          S         S
            C C C              S S S S         OO

**************************************************
     Computer Science is Cool Stuff!!!
```